Automated Face Analysis

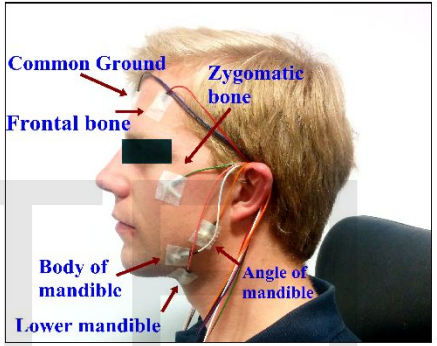
NPTEL

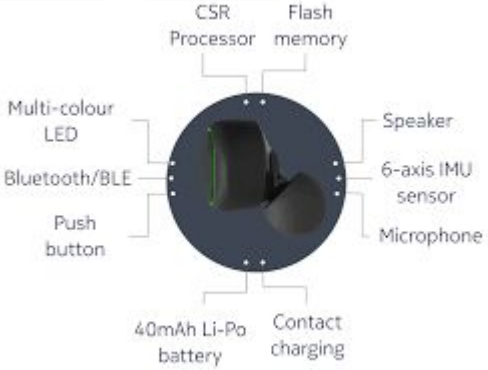
Dr. Jainendra Shukla
jainendra@iiitd.ac.in
B-501, R & D Block
Computer Science and Engineering

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Introduction

- FER
  - Facial emotion recognition
  - Facial expression recognition
- Inputs to FER:
  - Electromyograph (EMG)
  - Electrocardiogram (ECG),
  - Electroencephalograph (EEG),
  - Earables
  - Camera
    - Most informative
    - Non-intrusive





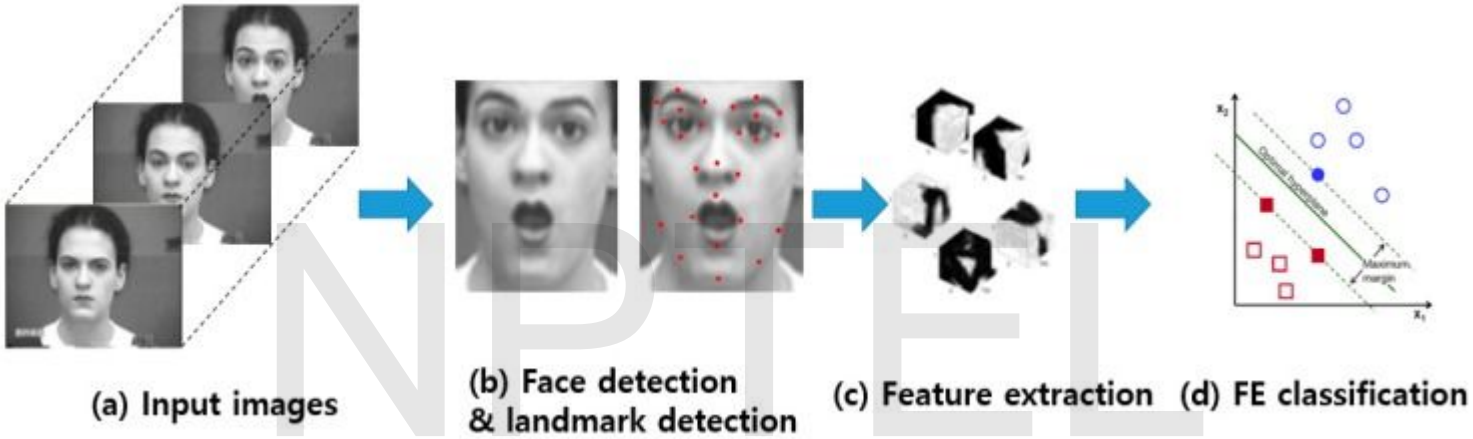
NPTEL

Static and Dynamic FER

- Static facial features obtained by extracting handcrafted features from selected peak expression frames of image sequences.
- Spatio-temporal features to capture the expression dynamics in facial expression sequences
- Recognition Rate
  - Dynamic FER has higher recognition rate than static FER because it provides additional temporal information.
- Drawbacks:
  - Extracted dynamic features have different transition durations and different feature characteristics of the facial expression depending on the particular faces.

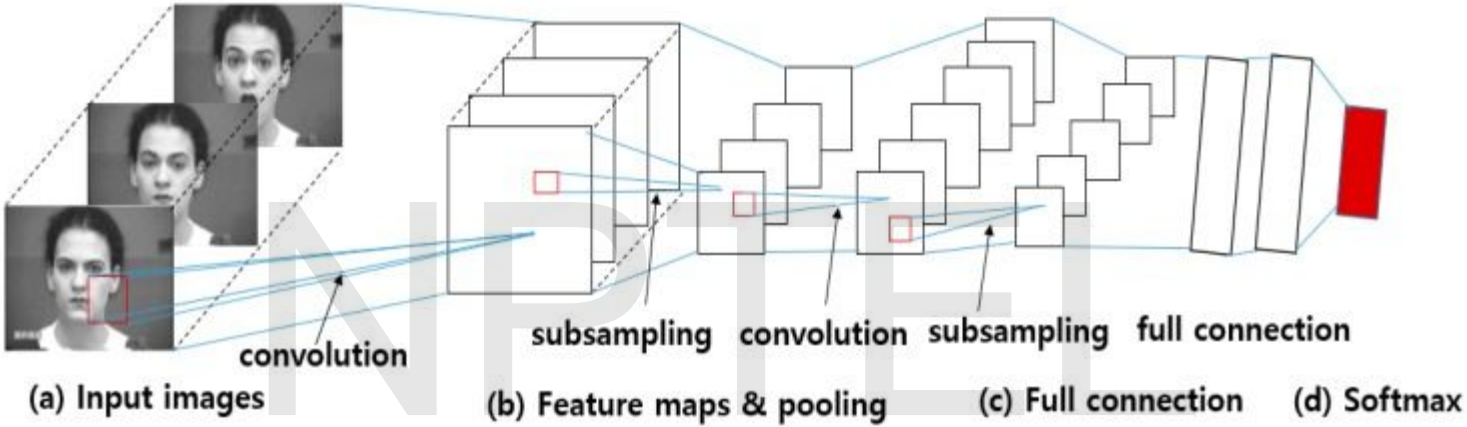
NPTEL

Conventional FER



- Three major steps:
  - Face and facial component detection,
  - Feature extraction, and
  - Expression classification

Deep-learning-based FER



- Deep-learning-based FER approaches:
  - highly reduce the dependence on face-physics-based models and other pre-processing techniques
  - Enabling “end-to-end” learning to occur in the pipeline directly from the input images

Macro Expressions

- Macro Expressions
  - Used primarily to capture Obvious or "normal" facial expressions.
  - Visually observed through Facial Landmarks which are salient points in facial regions such as the end of the nose, ends of the eye brows, and the mouth etc.
  - Last between 1/2 a second to 4 seconds
  - Match the content and tone of what is said

NPTEL

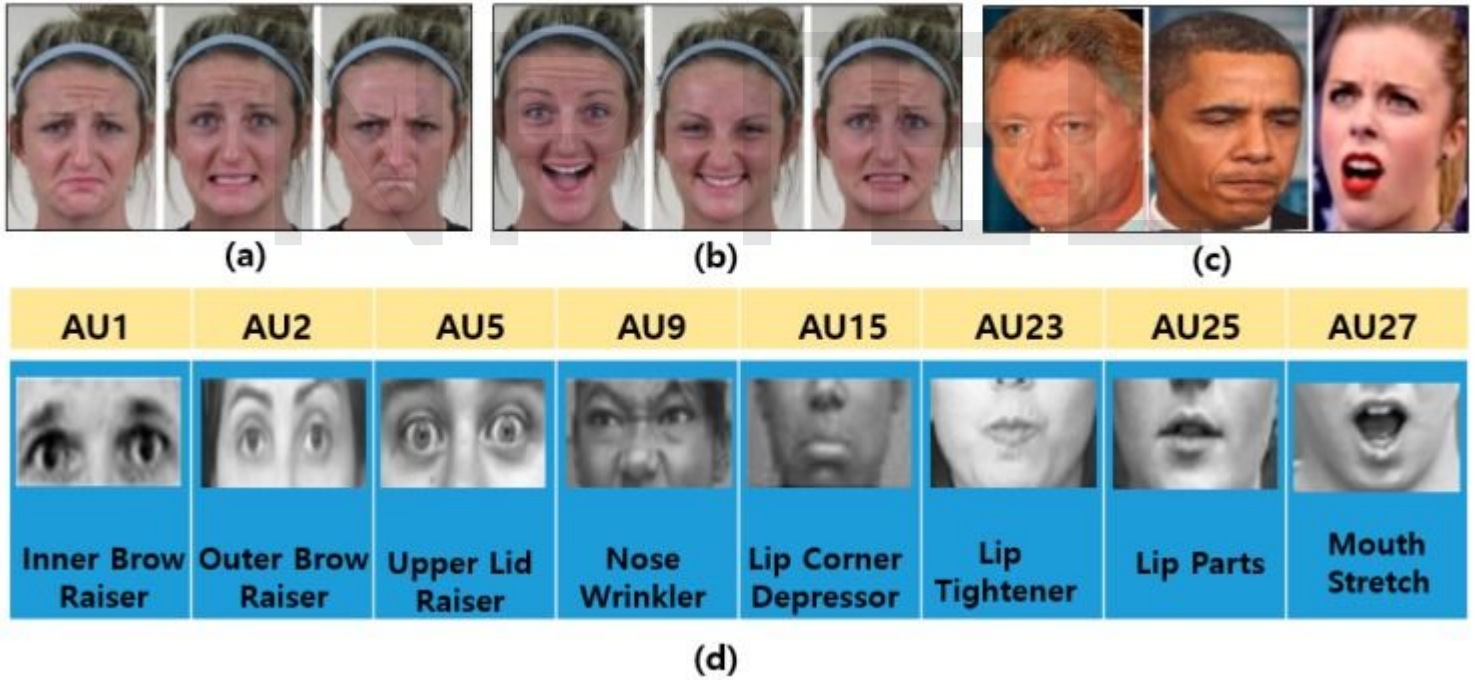
Micro Expressions

- Micro expressions (MEs)
  - More spontaneous and subtle facial movements that occur involuntarily. They tend to reveal a person's genuine and underlying emotions within a short period of time.
  - Often misinterpreted or missed altogether
  - Occur in 1/2 a second or less
  - Unconsciously display a concealed emotion

NPTI

Facial Action Coding System

- Facial action coding system (FACS)
  - A system based on facial muscle changes and can characterize facial actions to express individual human emotions (Ekman and Friesen , 1978).



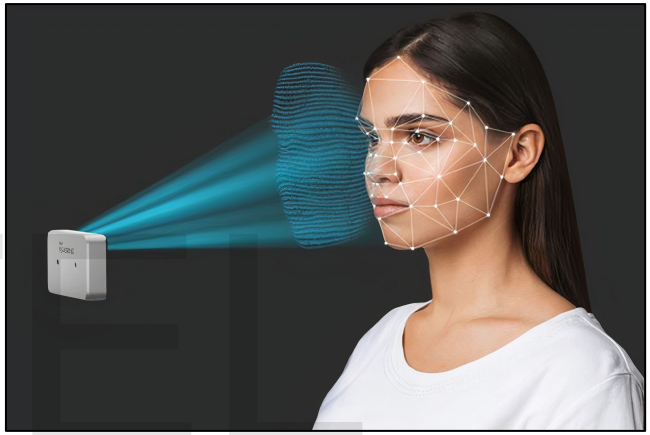
AUs (Benitez-Quiroz et al., 2017)

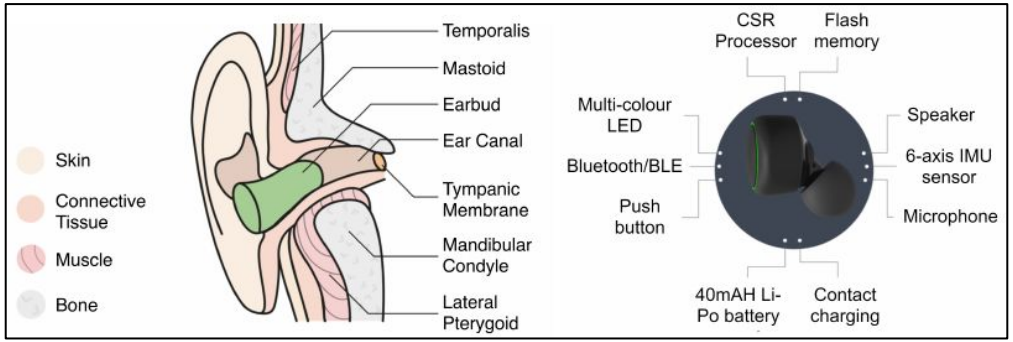
Category	AUs	Category	AUs
Happy	12, 25	Sadly disgusted	4, 10
Sad	4, 15	Fearfully angry	4, 20, 25
Fearful	1, 4, 20, 25	Fearfully surprised	1, 2, 5, 20, 25
Angry	4, 7, 24	Fearfully disgusted	1, 4, 10, 20, 25
Surprised	1, 2 , 25, 26	Angrily surprised	4, 25, 26
Disgusted	9, 10, 17	Disgusted surprised	1, 2, 5, 10
Happily sad	4, 6, 12, 25	Happily fearful	1, 2, 12, 25, 26
Happily surprised	1, 2, 12, 25	Angrily disgusted	4, 10, 17
Happily disgusted	10, 12, 25	Awed	1, 2, 5, 25
Sadly fearful	1, 4, 15, 25	Appalled	4, 9, 10
Sadly angry	4, 7, 15	Hatred	4, 7, 10
Sadly surprised	1, 4, 25, 26	-	-

NPTEL

ExpressEar: Sensing Facial Expressions

- Facial expressions: Common indicators of our mental state and emotions.
- Traditional Methods: Obtrusive | Occlusion variant | Privacy concerns
- *ExpressEar*, a novel FER system that repurposes commercial earables augmented with inertial sensors to capture fine-grained facial muscle movements.





Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, IMWUT' 21

Feature Extraction

- Geometric Features
  - The relationship between facial components is used to construct a feature vector for training.
  - (Ghimire and Lee, 2013) used two types of geometric features based on the position and angle of 52 facial landmark points, the angle and Euclidean distance between each pair of landmarks.

- Shape or geometric features alone are insufficient for some AUs
  - AU6 and AU7 both narrow the eye aperture

- Appearance or texture information may aid in discriminating between them.
  - E.g. : AU6 but not AU7 causes wrinkles lateral to the eye corners.

NPTEL

Feature Extraction

- Appearance Features
  - Represent changes in skin texture such as wrinkling and deepening of facial furrows and pouching of the skin.

- Many techniques for describing facial texture have been proposed.
  - Simplest is a raw pixel-intensity values.
    - Prone to lightning conditions

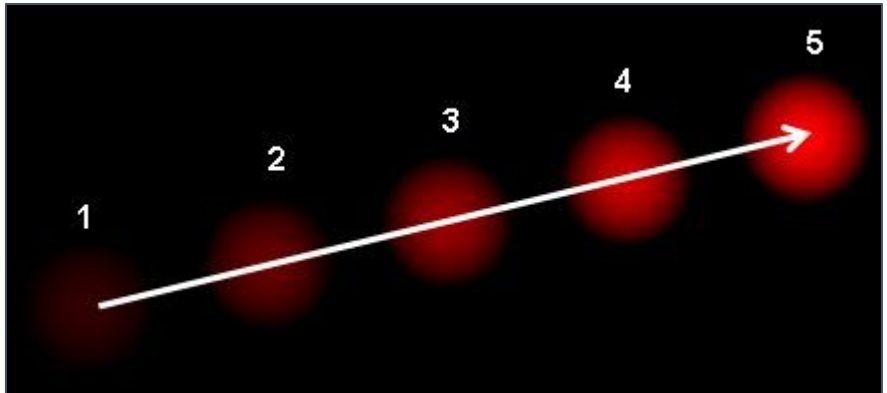
- Robust Appearance features
  - Gabor wavelets or magnitudes (Jones & Palmer, 1987)
  - Histogram of oriented gradients (HOG) (Dalal & Triggs, 2005)
  - SIFT (Mikolajczyk & Schmid, 2005)

NPTEL

Feature Extraction

- Motion Features
  - Motions is an important cue to expression recognition, especially for subtle expressions (Ambadar et.a l., 2005).

- Motion features include:
  - Optical Flow (Mase, 1991)
  - Motion history images (MHI) (Chetverikov & Peteri, 2005).
  - Local binary patterns (LBPs) (Zhao & Piettkainen, 2007)



NPTEL

FER Databases

- The Extended Cohn-Kanade Dataset (CK+):
  - (Lucey P., Cohn J.F., Kanade T., Saragih J., Ambadar Z., Matthews I, 2010)
  - CK+ contains 593 video sequences on both posed and non-posed (spontaneous) emotions, along with additional types of metadata.
  - The age range of its 123 subjects is from 18 to 30 years, most of who are female
  - Each video shows a facial shift from the neutral expression to a targeted peak expression.

NPTEL

FER Databases

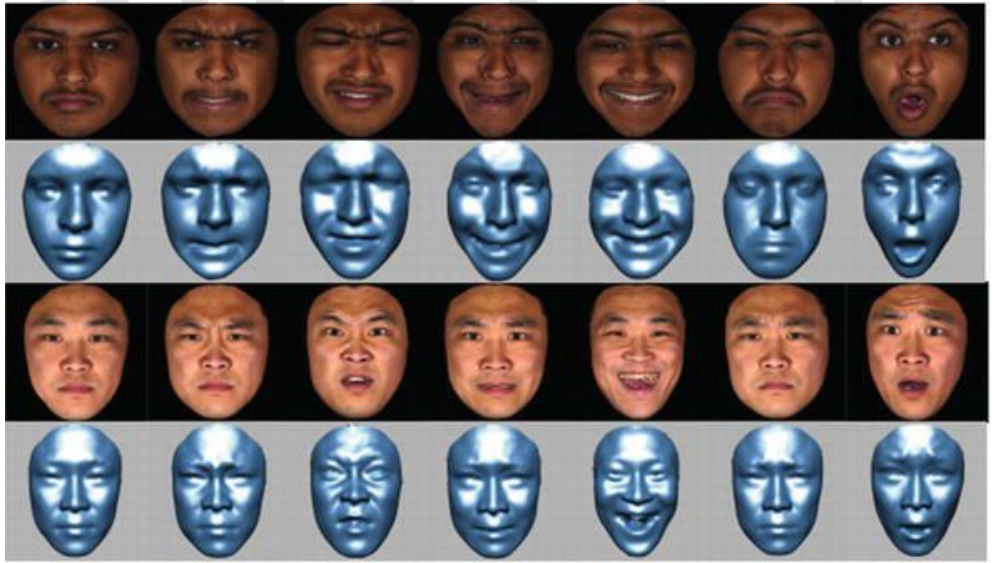
- Compound Emotion (CE)
  - (Tao S.Y., Martinez A.M., 2014)
  - CE contains 5060 images corresponding to 22 categories of basic and compound emotions for its 230 human subjects (130 females and 100 males, mean age of 23)

- Denver Intensity of Spontaneous Facial Action Database (DISFA)
  - (Mavadati S.M., Mahoor M.H., Bartlett K., Trinh P., Cohn J., 2013)
  - DISFA consists of 130,000 stereo video frames at high resolution (1024 × 768) of 27 adult subjects (12 females and 15 with different ethnicities.

NPTEL

FER Databases

- Binghamton University 3D Facial Expression
  - (Lijun Yin; et. al., 2008)
  - 100 subjects (56% female, 44% male), ranging age from 18-70 years old
  - Each subject performed seven expressions in front of the 3D face scanner.
  - Each expression sequence contains about 100 frames.



Applications

- Physical Pain
- Depression and Psychological Distress
- Deception Detection
- Drowsy Driver Detection
- In-class attention?
- Use of additional modalities:
  - E.g. - Depression: Voice is more prominent

NPTEL

Limitations

- Real-time systems: Computational complexity
- Illumination
- Occlusion
  - Self-occlusion
  - External occlusions
- Subtle facial expression
- Individual variability
- Do we need better features and classifiers or more data?
  - Both may be necessary
- Ethical Issues:
  - Fairness
  - Accountability
  - Transparency

NPTEL

References

1. P. Ekman and W. Friesen, “Constants across cultures in the face and emotion.,” Journal of Personality and Social Psychology, vol. 17, pp. 124–129, 1971.
2. Benitez-Quiroz C.F., Srinivasan R., Martinez A.M. EmotioNet: An accurate, real-time algorithm for the automatic annotation of a million facial expressions in the wild; Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition; Las Vegas, NV, USA. 26 June–1 July 2016; pp. 5562–5570.
3. FACS: https://imotions.com/blog/facial-action-coding-system/
4. Ghimire D., Lee J. Geometric feature-based facial expression recognition in image sequences using multi-class AdaBoost and support vector machines. Sensors. 2013;13:7714–7734. doi: 10.3390/s130607714.

NPTEL

References

1. Feature Extraction: https://www.cs.cmu.edu/~cga/behavior/FEA-Bookchapter.pdf
2. Lucey P., Cohn J.F., Kanade T., Saragih J., Ambadar Z., Matthews I. The extended Cohn-Kanade Dataset (CK+): A complete dataset for action unit and emotion-specified expression; Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops; San Francisco, CA, USA. 13–18 June 2010; pp. 94–101.
3. Tao S.Y., Martinez A.M. Compound facial expressions of emotion. Natl. Acad. Sci. 2014;111:E1454–E1462.
4. Mavadati S.M., Mahoor M.H., Bartlett K., Trinh P., Cohn J. DISFA: A spontaneous facial action intensity database. IEEE Trans. Affect. Comput. 2013;4:151–160. doi: 10.1109/T-AFFC.2013.4.

NPTEL